SSR-NET: Spatial–Spectral Reconstruction Network for Hyperspectral and Multispectral Image Fusion

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***Abstract*— The fusion of a low-spatial-resolution hyperspectral image (HSI) (LR-HSI) with its corresponding high-spatial-resolution multispectral image (MSI) (HR-MSI) to reconstruct a high-spatial-resolution HSI (HR-HSI) has been a significant subject in recent years. Nevertheless, it is still difficult to achieve the cross-mode information fusion of spatial mode and spectral mode when reconstructing HR-HSI for the existing methods. In this article, based on a convolutional neural network (CNN), an interpretable spatial–spectral reconstruction network (SSR-NET) is proposed for more efficient HSI and MSI fusion. More specifically, the proposed SSR-NET is a physical straightforward model that consists of three components: 1) cross-mode message inserting (CMMI); this operation can produce the preliminary fused HR-HSI, preserving the most valuable information of LR-HSI and HR-MSI; 2) spatial reconstruction network (SpatRN); the SpatRN concentrates on reconstructing the lost spatial information of LR-HSI with the guidance of spatial edge loss ($\mathcal{L}_{\text{spat}}$); and 3) spectral reconstruction network (SpecRN); the SpecRN pays attention to reconstruct the lost spectral information of HR-MSI under the constraint of spatial edge loss ($\mathcal{L}_{\text{spec}}$). Comparative experiments are conducted on six HSI data sets of Urban, Pavia University (PU), Pavia Center (PC), Botswana, Indian Pines (IP), and Washington DC Mall (WDCM), and the proposed SSR-NET achieves the superior or competitive results in comparison with seven state-of-the-art methods. The code of SSR-NET is available at https://github.com/hw2hwei/SSRNET.**

***Index Terms*— Convolutional neural network (CNN), cross-mode message inserting (CMMI), hyperspectral image (HSI), image fusion, multispectral image (MSI), spatial–spectral reconstruction network (SSR-NET).**

I. Introduction

HYPERSPECTRAL imaging is a technology where images of hundreds of narrow spectral bands with different wavelengths can be obtained. Since hyperspectral images (HSIs) have a high spectral coverage that can accurately identify the materials and objects on the ground, there are wide applications of HSI in the fields of image classification [1]–[3], object detection [4], band selection [5]–[8], change detection [9]–[11], and so on. However, long exposures of hyperspectral systems are necessary for enough signal-to-noise ratio (SNR), which leads to the low-spatial-resolution of HSIs (LR-HSIs). In contrast, multispectral systems can acquire high spatial resolution of multispectral images (MSIs) (HR-MSI). Thus, it is meaningful to reconstruct high-spatial-resolution HSIs (HR-HSI) with LR-HSI and HR-MSI, which is called as HSI and MSI fusion.

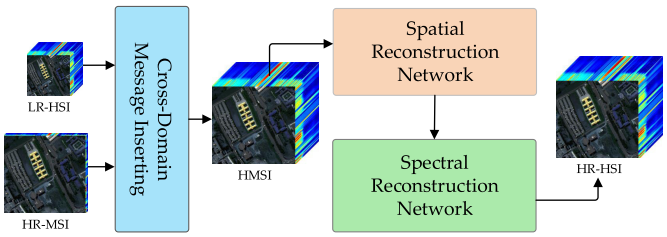


Fig. 1. Reconstructing HR-HSI from the corresponding HR-MSI and LR-HSI. First, the messages of HR-MSI and LR-HSI are passed across spatial and spectral modes. Then, in the orange stage, the SpatRN aims to reconstruct the lost spatial information of LR-HSI. Finally, in the green stage, the SpecRN aims to reconstruct the lost spectral information based on the HSI, which is reconstructed in spatial mode, and generates the estimated HR-HSI.

In recent years, a number of studies have been done in the field of HSI and MSI fusion [12]–[17], which can be roughly divided into two categories: traditional methods and deep learning methods. In traditional methods, there are some different approaches, including matrix factorization-based methods, Bayesian-based methods, and tensor-based methods. Although these methods have achieved excellent performance on LR-HSI and HR-MSI fusion, it is still challenging to efficiently pass messages across spatial and spectral modes, which is crucial for improving the fusion quality.

Compared with traditional methods, methods based on deep learning, especially the convolutional neural network (CNN) [18]–[20], have shown superior performance due to their powerful feature-extraction ability. Hence, based on CNN, as shown in Fig. 1, a spatial–spectral reconstruction network (SSR-NET) is proposed for LR-HSI and HR-MSI fusion for the first time in this article. Different from the previous work, the proposed method focuses on passing valuable messages across spatial and spectral modes in an efficient and interpretable manner. More concretely, the proposed SSR-NET is a three-stage network model made up of three modules: 1) cross-mode message inserting (CMMI); 2) spatial reconstruction

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network (SpatRN) with the spatial edge loss optimized by $\mathcal{L}_{\text{spat}}$; and 3) spectral reconstruction network (SpecRN) with the spectral edge loss optimized by $\mathcal{L}_{\text{spec}}$.

First, as the input of SSR-NET, the LR-HSI and HR-MSI are sent into the CMMI, which maintains the valuable spectral information of LR-HSI and spatial information of HR-MSI and then fuses them into a hypermultiple spectral image (HMSI) that has the same size as the reference HR-HSI (i.e., the ground-truth HR-HSI). Second, the SpatRN is designed to reconstruct the lost spatial information of HMSI. However, CNN is a black-box model, and the learned features are lack of enough explanation. To assign the physical meaning to SpatRN, the spatial edge loss of $\mathcal{L}_{\text{spat}}$ is presented to constrain SpatRN to focus on spatial reconstruction. Finally, the SpecRN is designed to reconstruct the lost spectral information from the reconstructed HMSI in spatial mode. Similarly, the spectral edge loss of $\mathcal{L}_{\text{spec}}$ is designed to make SpecRN pay attention to spectral restoration.

In summary, the main contributions of this article conclude the following aspects.

1) Based on CNN, a novel SSR-NET is first proposed for more efficient HSI and MSI fusion in this article.
2) The proposed SSR-NET is a physical straightforward CNN model, which is under the constraint of spatial edge loss $\mathcal{L}_{\text{spat}}$ and spectral edge loss $\mathcal{L}_{\text{spec}}$. $\mathcal{L}_{\text{spat}}$ and $\mathcal{L}_{\text{spec}}$ are specially designed for spatial and spectral reconstruction.
3) Compared with seven state-of-the-art approaches, the proposed SSR-NET achieves the best results on five HSI data sets of Urban, Pavia University (PU), Pavia Center (PC), Botswana and Indian Pines (IP), and the competitive results on the data set of Washington DC Mall (WDCM). Such experimental results demonstrate the effectiveness and superiority of the proposed SSR-NET in LR-HSI and HR-MSI fusion.

The remainder of this article is organized as follows. Section II gives an introduction of the related works of LR-HSI and HR-MSI fusion. In Section III, we describe the proposed SSR-NET model in detail. The experimental results on four data sets are analyzed in Section IV. In the end, conclusions are provided in Section V.

II. Related Work

In this section, some classical methods of HSI and MSI fusion will be reviewed, which roughly fall into two categories: traditional methods and deep learning methods.

A. Traditional Methods

Generally, traditional methods include the following three types of methods.

1) Matrix Factorization-Based Methods: This type of methods usually unfold the 3-D HSI, where the three dimensions, respectively, denote the width, height, and the number of spectral bands, into a 2-D matrix, where the two dimensions denote the flattened spatial locations and the band number. In these methods, two matrices of an endmember matrix and an abundance matrix are expected to be, respectively, estimated from the LR-HSI and the HR-MSI and are used to reconstruct the corresponding HR-HSI.

For example, based on unsupervised unmixing, Yokoya *et al.* [12] presented a coupled nonnegative matrix factorization (CNMF) method for LR-HSI and HR-MSI fusion, where the abundance matrix of high spatial resolution acquired from MSIs and the hyperspectral endmember matrix are integrated to generate a new HR-HSI. Then, Lanaras *et al.* [13] proposed a method that concentrates on jointly the spectral unmixing problem for both two input images, where some constraints are added according to the physical properties of spectral unmixing. Dong *et al.* [14] presented a sparsity-based HSI super-resolution method on the basis of dictionary learning and sparse representation. After that, considering the ill-posed inverse problem, an ADMM-based CO-CNMF algorithm is proposed by Lin *et al.* [21] for further optimization of CNMF, where ℓ_1-norm and SSD regularizers are incorporated and added to CNMF criterion.

2) Bayesian-Based Methods: This type of method generally utilizes the appropriate prior distribution of the given images to solve the fusion problem of LR-HSI and HR-MSI.

For instance, a Bayesian sparse representation-based approach is first proposed by Akhtar *et al.* [15]. In this work, the nonparametric Bayesian dictionary learning is utilized to learn the distributions of the scene spectrum and the proportion of them in the image, where the distributions would be utilized to compute sparse codes of the high-resolution image. Wei *et al.* [22] proposed a fast multiband image fusion algorithm (FUSE) that is based on a solution of a Sylvester equation. Combining the alternating direction method of multiplier and block coordinate descent, this algorithm can be easily extended to calculate the Bayesian estimators of fusion.

3) Tensor Factorization-Based Methods: Different from matrix factorization-based methods, methods based on tensor factorization typically treat the HSI as a 3-D tensor, which has three modes as described earlier. In these methods, the HR-HSI is cut apart into some cubes, and then, the similar cubes would be grouped based on the learned clusters and some certain priors.

For example, on the basis of Tucker factorization, Dian *et al.* [16] first proposed a novel HSI superresolution method that is called the NLSTF. It unifies the sparse tensor factorization and the nonlocal means approach into one framework, thus considering the HSI superresolution problem from the point of the estimation of dictionaries and a sparse core tensor for each cube. Similarly, a coupled sparse tensor factorization-based framework named CSTF is proposed in [23], where the estimation of the core tensor and three dictionaries are formulated as a coupled sparse tensor decomposition of the HR-MSI and LR-HSI. In [24], taking geometric structures into account, a spatial–spectral-graph-regularized low-rank tensor decomposition method (SSGLRTD) is presented by Zhang *et al.* [24]. Meanwhile, an ALM-based algorithm is specially designed for better optimization of the fusion model. Following the previous work in [16], Dian *et al.* [25] designed a novel LTTR prior to learning the relationship among the spectral, spatial, and nonlocal modes of the nonlocal similar HR-HSI cubes. In this work, the similar HR-HSI

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cubes are composed into a 4-D tensor, and the ADMM [26] algorithm is adopted to solve the optimization problem.

B. Deep Learning Methods

With the rapid development of deep learning methods, especially CNN [18]–[20], [27]–[29], these types of methods have become a growing trend in all kinds of HSI processing [30]–[32]. In LR-HSI and HR-MSI fusion, deep learning methods show excellent performance. Different from traditional methods, in deep learning-based approaches, various neural networks are usually exploited to enhance the performance of image fusion in a learnable manner.

For example, Palsson *et al.* [33] first proposed a 3-D-CNN to acquire HR-HSI from LR-HSI and HR-MSI, where PCA prior is utilized for dimensionality reduction of the fusion. After that, a deep HSI sharpening method (DHSIS) is proposed by Dian *et al.* [17], which learns the priors by residual learning to achieve the regularization of image fusion problem. A novel MS/HS fusion network is presented by Xie *et al.* [34], which considers the generation mechanism underlying the MSI/HSI fusion data. Following that, based on CNN denoiser and subspace representation, Dian *et al.* [35] proposed a new HSI-MSI fusion method that can be applied to different HSI data sets without retraining. By utilizing techniques of the network-in-network convolutional unit, skip connection, and batch normalization, Xu *et al.* [36] proposed a two-branch network (HAM-MFN) to gradually reconstruct HR-HSI via fusing LR-HSI and HR-MSI at different scales, where a RAP loss is designed to deal with spectral and spatial distortions. Some other related deep learning methods, including TFNet [37], ResTFNet [37], SSFCNN [38], ConSSFCNN [38], and MSDCNN [39], are introduced in detail in Section IV as the comparative methods.

III. Methodology

In this section, the proposed SSR-NET, which is shown in Fig. 2, will be introduced in detail. Overall, the proposed SSR-NET is a physical straightforward CNN model, and it mainly consists of three modules: 1) CMMI; 2) SpatRN with spatial edge loss ($\mathcal{L}_{\text{spat}}$); and 3) SpecRN with spectral edge loss ($\mathcal{L}_{\text{spec}}$).

A. Cross-Mode Message Inserting

In the proposed SSR-NET, the reference HR-HSI and the estimated HR-HSI are denoted as $\mathbf{R} \in \mathbb{R}^{H\times W\times L}$ and $\mathbf{Z} \in \mathbb{R}^{H\times W\times L}$, where H and W, respectively, represent the dimensions of the height and width, and L represents the number of spectral bands. Besides, its inputs are a LR-HSI denoted as $\mathbf{X} \in \mathbb{R}^{h\times w\times L}$ ($h \ll H, w \ll W$) and a HR-MSI denoted as $\mathbf{Y} \in \mathbb{R}^{H\times W\times l}$ ($l \ll L$). $\mathbf{X}$ and $\mathbf{Y}$ are, respectively, sampled in the spatial and spectral modes, which are obtained by

$$\begin{cases} \mathbf{X} = \text{Gaussian}(\mathbf{Z}) \\ \mathbf{X} = \text{Bilinear}(\mathbf{X}, 1/r \end{cases} \tag{1}$$

$$\begin{cases} \mathbf{Y}(k) = \mathbf{Z}(s_k), k \in \{1, \ldots, l\} \\ s_k = (k-1) * L/(l-1), s_k \in \{s_1, \ldots, s_l\} \end{cases} \tag{2}$$

where $\mathbf{X}$ is spatially downsampled by bilinear operation at the ratio of r from $\mathbf{Z}$, which is blurred by Gaussian filter in advance. $\mathbf{Y}$ is sampled from $\mathbf{Z}$ at an equal interval of bands. $\mathbf{Y}(k)$ represents the kth band of $\mathbf{Y}$, and $\{s_1, \ldots, s_l\}$ represent the sampled band numbers in HR-HSI.

The goal of CMMI is to produce a preliminary concatenated hypermultiple spectral image (HMSI) represented as $\mathbf{Z}_{\text{pre}} \in \mathbb{R}^{H\times W\times L}$, which takes advantages of both the spatial information of HR-MSI and the spectral information of LR-HSI preserving their relative spatial–spectral position.

By utilizing bilinear interpolation, the LR-HSI of $\mathbf{X}$ would be upsampled to the same size as the HR-MSI of $\mathbf{Y}$ in the spatial mode, which is denoted as

$$\mathbf{X}\uparrow = \text{Bilinear}(\mathbf{X}, r) \tag{3}$$

where r is the upsampling ratio, and $\mathbf{X}\uparrow$ is the upsampled LR-HSI.

Then, the HR-MSI and the upsampled LR-HSI would be preliminarily fused, which is formulated as

$$\mathbf{Z}_{\text{pre}}(k) = \begin{cases} \mathbf{Y}(k), & \text{if } k \in \{s_1, \ldots, s_l\} \\ \mathbf{X}(k)\uparrow, & \text{otherwise} \end{cases} \tag{4}$$

where $\mathbf{Z}_{\text{pre}}$ denotes the HMSI, and $\mathbf{Z}_{\text{pre}}(k)$ is the kth band of HMSI. Similarly, $\mathbf{Y}(k)$ and $\mathbf{X}(k)\uparrow$ represent the kth band of HR-MSI and the upsampled LR-HSI, respectively. In this way, the concatenated HMSI can contain both spatial and spectral information of HR-MSI and LR-HSI.

To preliminarily pass the information between the modes of spatial and spectral, a convolutional layer with the kernel size set to 3×3 and spatial height and width stride set to 1 is applied in HMSI. It is denoted as

$$\mathbf{Z}_{\text{pre}} = \text{ReLU}(\text{Conv}_{\text{pre}}(\mathbf{Z}_{\text{pre}})) \tag{5}$$

where ReLU is the nonlinear activation function of rectified linear unit [40].

B. Spatial Reconstruction Network With Spatial Edge Loss

In order to reconstruct the spatial information from $\mathbf{Z}_{\text{pre}}$, another 3×3 convolutional layer with height/width stride set to 1 serves as the SpatRN, which is formulated as

$$\mathbf{Z}_{\text{spat}} = \mathbf{Z}_{\text{pre}} + \text{Conv}_{\text{spat}}(\mathbf{Z}_{\text{pre}}) \tag{6}$$

where $\text{Conv}_{\text{spat}}$ represents the convolutional layer. It is worth mentioning that the skip-connection operation is used to improve the model stability during the training stage.

However, due to the black-box characteristic of CNN, it is uncontrollable for the learned feature mapping. As we all know, the spatial edges of images contain the high-frequency feature, which are crucial for spatial reconstruction. To make SpatRN focus on the restoration of spatial information, based on spatial edges, as shown in Fig. 3, a novel spatial edge loss ($\mathcal{L}_{\text{spat}}$) is proposed in this article to constrain the output of

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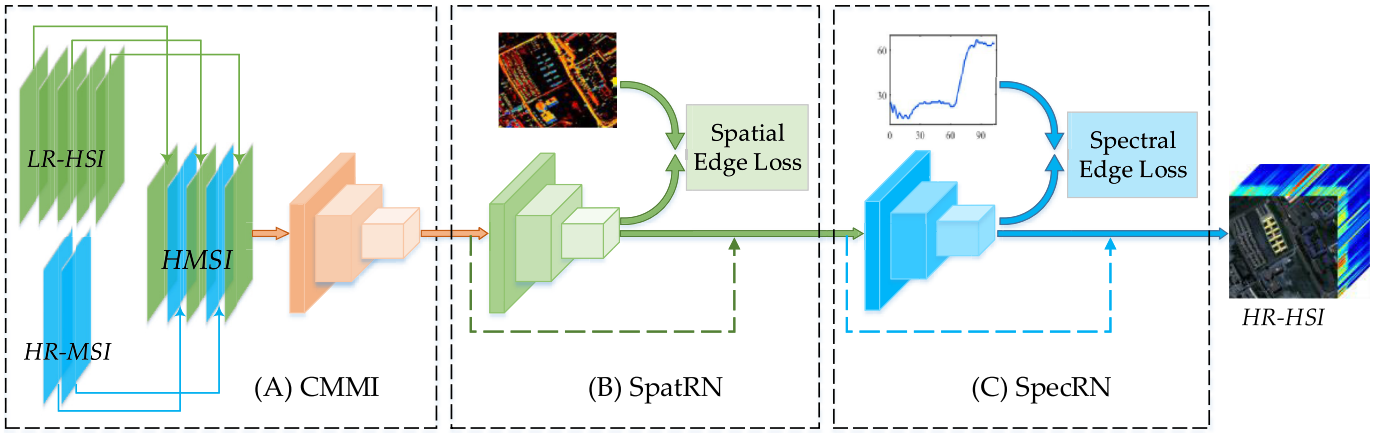


Fig. 2. Framework of the proposed SSR-NET. (a) CMMI. (b) SpatRN. (c) SpecRN.

SpatRN. $\mathcal{L}_{\text{spat}}$ is calculated by

$$\mathbf{E}_{\text{spat1}}(i,j,k) = \mathbf{Z}_{\text{spat}}(i+1,j,k) - \mathbf{Z}_{\text{spat}}(i,j,k) \tag{7}$$

$$\mathbf{E}_{\text{spat2}}(i,j,k) = \mathbf{Z}_{\text{spat}}(i,j+1,k) - \mathbf{Z}_{\text{spat}}(i,j,k) \tag{8}$$

$$\bar{\mathbf{E}}_{\text{spat1}}(i,j,k) = \mathbf{Z}(i+1,j,k) - \mathbf{Z}(i,j,k) \tag{9}$$

$$\bar{\mathbf{E}}_{\text{spat2}}(i,j,k) = \mathbf{Z}(i,j+1,k) - \mathbf{Z}(i,j,k) \tag{10}$$

$$\mathcal{L}_{\text{spat1}} = \frac{\sum_{k=1}^{L}\sum_{i=1}^{H-1}\sum_{j=1}^{W}(\mathbf{E}_{\text{spat1}}(i,j,k) - \bar{\mathbf{E}}_{\text{spat1}}(i,j,k))^2}{2WL(H-1)} \tag{11}$$

$$\mathcal{L}_{\text{spat1}} = \frac{\sum_{k=1}^{L}\sum_{i=1}^{H}\sum_{j=1}^{W-1}(\mathbf{E}_{\text{spat2}}(i,j,k) - \bar{\mathbf{E}}_{\text{spat2}}(i,j,k))^2}{2HL(W-1)} \tag{12}$$

$$\mathcal{L}_{\text{spat}} = 0.5 * \mathcal{L}_{\text{spat1}} + 0.5 * \mathcal{L}_{\text{spat2}} \tag{13}$$

where $\mathbf{E}_{\text{spat1}} \in \mathbb{R}^{(H-1)\times W\times L}$ and $\mathbf{E}_{\text{spat2}} \in \mathbb{R}^{H\times(W-1)\times L}$ are the edge maps of $\mathbf{Z}_{\text{spat}}$ along spatial height and width, respectively. Similarly, $\bar{\mathbf{E}}_{\text{spat1}} \in \mathbb{R}^{(H-1)\times W\times L}$ and $\bar{\mathbf{E}}_{\text{spat2}} \in \mathbb{R}^{H\times(W-1)\times L}$ are the edge maps of the reference HR-HSI of $\mathbf{Z}_{\text{spat}}$ along spatial height and width, respectively. $\mathcal{L}_{\text{spat1}}$ and $\mathcal{L}_{\text{spat2}}$ are the spatial height edge loss and spatial width edge loss, which are calculated between $\mathbf{E}_{\text{spat}}$ and $\bar{\mathbf{E}}_{\text{spat}}$ by the loss function of mean squared error (MSE). Finally, $\mathcal{L}_{\text{spat1}}$ and $\mathcal{L}_{\text{spat2}}$ are fused into $\mathcal{L}_{\text{spat}}$.

C. Spectral Reconstruction Network With Spectral Edge Loss

After spatial reconstruction, one more convolutional layer serving as the SpecRN with the same architecture as SpatRN is used to further reconstruct the spectral information based on $\mathbf{Z}_{\text{spat}}$. It is formulated as

$$\mathbf{Z}_{\text{spec}} = \mathbf{Z}_{\text{spat}} + \text{Conv}_{\text{spec}}(\mathbf{Z}_{\text{spat}}) \tag{14}$$

where $\text{Conv}_{\text{spec}}$ represents the convolutional layer. The skip-connection operation is also applied in SpatRN.

Similar to spatial edges, as shown in Fig. 3, the spectral edges of bands contain the high-frequency information that is crucial for spectral reconstruction. To make SpecRN pay attention to spectral restoration, the spectral edge loss ($\mathcal{L}_{\text{spec}}$) is also proposed in this article to constrain the output of SpecRN. $\mathcal{L}_{\text{spec}}$ is calculated by

$$\mathbf{E}_{\text{spec}}(i,j,k) = \mathbf{Z}_{\text{spec}}(i,j,k+1) - \mathbf{Z}_{\text{spec}}(i,j,k) \tag{15}$$

$$\bar{\mathbf{E}}_{\text{spec}}(i,j,k) = \mathbf{Z}(i,j,k+1) - \mathbf{Z}(i,j,k) \tag{16}$$

$$\mathcal{L}_{\text{spec}} = \frac{\sum_{k=1}^{L}\sum_{i=1}^{H-1}\sum_{j=1}^{W}(\mathbf{E}_{\text{spec}}(i,j,k) - \bar{\mathbf{E}}_{\text{spec}}(i,j,k))^2}{2HW(L-1)} \tag{17}$$

where $\mathbf{E}_{\text{spec}} \in \mathbb{R}^{H\times W\times(L-1)}$ and $\bar{\mathbf{E}}_{\text{spec}} \in \mathbb{R}^{H\times W\times(L-1)}$ are the edge maps of $\mathbf{Z}_{\text{spec}}$ and $\mathbf{Z}$ along the spectral mode, respectively. The spectral edge loss of $\mathcal{L}_{\text{spec}}$ is the MSE loss of $\mathbf{E}_{\text{spec}}$ and $\bar{\mathbf{E}}_{\text{spec}}$. After CMMI, SpatRN, and SpecRN, $\mathbf{Z}_{\text{spec}}$ is used as the final estimated HR-HSI denoted as $\mathbf{Z}_{\text{fus}}$. It is formulated as

$$\mathbf{Z}_{\text{fus}} = \mathbf{Z}_{\text{spec}}. \tag{18}$$

For $\mathbf{Z}_{\text{fus}}$, it is optimized by the fusion loss denoted as $\mathcal{L}_{\text{spec}}$, which is formulated as

$$\mathcal{L}_{\text{fus}} = \frac{\sum_{k=1}^{L}\sum_{i=1}^{H}\sum_{j=1}^{W}(\mathbf{Z}_{\text{fus}}(i,j,k) - \bar{\mathbf{Z}}(i,j,k))^2}{2WHL}. \tag{19}$$

In the proposed SSR-NET, the overall loss denoted as $\mathcal{L}$ is the sum of $\mathcal{L}_{\text{spat}}$, $\mathcal{L}_{\text{spec}}$, and $\mathcal{L}_{\text{fus}}$

$$\mathcal{L} = \mathcal{L}_{\text{spat}} + \mathcal{L}_{\text{spec}} + \mathcal{L}_{\text{fus}}. \tag{20}$$

D. Variations of SSR-NET

To explore the effectiveness of the components of SSR-NET, including CMMI, SpatRN, and SpecRN, five variant models of Spat-CNN, Spec-CNN, SpatR-NET, SpecR-NET, and SSR-NET are designed for ablation experiments.

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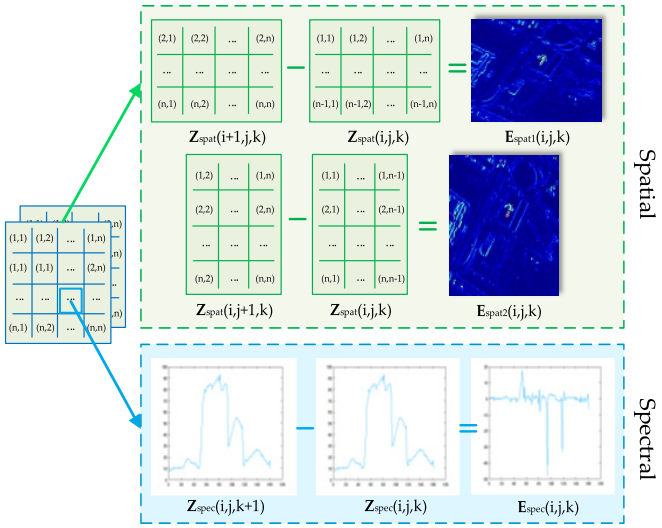


Fig. 3. Illustration of spatial and spectral edges.

TABLE I

ARCHITECTURE AND OPTIMIZATION LOSS OF THE VARIANT MODELS OF THE PROPOSED SSR-NET ON THE URBAN DATA SET

Model	Input	Architecture	Loss
Spat-CNN	LR-HSI ($\mathbf{X}$)	Conv(3×3, L, L)	
		Conv(3×3, L, L)	$\mathcal{L}_{fus}$
Spec-CNN	HR-MSI ($\mathbf{Y}$)	Conv(3×3, l, L)	
		Conv(3×3, L, L)	$\mathcal{L}_{fus}$
SpatR-NET	HMSI ($\mathbf{Z}_{pre}$)	Conv(3×3, L, L)	$\mathcal{L}_{spat}$
		Conv(3×3, L, L)	$\mathcal{L}_{fus}$
SpecR-NET	HMSI ($\mathbf{Z}_{pre}$)	Conv(3×3, l, L)	$\mathcal{L}_{spec}$
		Conv(3×3, L, L)	$\mathcal{L}_{fus}$
SSR-NET	HMSI ($\mathbf{Z}_{pre}$)	Conv(3×3, L, L)	
		Conv(3×3, L, L)	$\mathcal{L}_{spat}$
		Conv(3×3, L, L)	$\mathcal{L}_{spec}$+$\mathcal{L}_{fus}$

The architectures and the layerwise loss of these five models are provided in Table I. Under the constraints of fusion loss of $\mathcal{L}_{\text{fus}}$, Spat-CNN and Spec-CNN aim to reconstruct the HR-HSI using LR-HSI and HR-MSI, respectively. In contrast, the inputs of SpatR-NET, SpecR-NET, and SSR-NET are all the HMSI. The SpatR-NET focuses on restoring the spatial information of the HMSI with the guidance of $\mathcal{L}_{\text{fus}}$ and the spatial edge loss of $\mathcal{L}_{\text{spat}}$, while the SpatR-NET focuses on restoring the spectral information of the HMSI with the guidance of $\mathcal{L}_{\text{fus}}$ and the spectral edge loss of $\mathcal{L}_{\text{spec}}$. SSR-NET is a combination of SpatR-NET and SpecR-NET. In this table, Conv(3×3, l, L) denotes the 3×3 convolutional layer, whose input and output channel numbers are, respectively, l and L with the height/width stride set to 1. Besides, each convolutional layer is followed by the nonlinear activation function of ReLU.

IV. EXPERIMENTS

In this section, sufficient experiments are conducted on six data sets to verify the effectiveness of the proposed SSR-NET. First, the experimental data sets and evaluation metrics are introduced. Then, the experimental settings are provided in detail. Following that, the ablation experiments of SSR-NET are performed to explore the role of its components. Moreover, the comparison experiments of SSR-NET and some state-of-the-art methods are conducted. Finally, we quantitatively analyze the interpretability of SSR-NET and make a comparison between SSR-NET and some other deep learning methods in model size and test time.

A. Data Sets

In this article, six data sets are adopted to verify the effectiveness and generalization ability of the proposed SSR-NET, including Urban, PU, PC, Botswana, IP, and Washington DC Mall (WDCM).

1) Urban: The HYDICE Urban data set was obtained over Copperas Cove, TX, USA, in 1995. In this data set, there are 210 bands in total that cover the wavelengths from 0.4 to 2.5 μm with an interval of 10 nm. After removing the bands of dense water vapor and atmospheric, the remaining 162 bands are used in this article, and the image of each band measures 307×307 pixels with a spatial resolution of 2 m.

2) Pavia University: The PU data set was obtained by the reflective optics spectrographic imaging system (ROSIS) sensor over PU, Italy, in 2003. In this data set, there are 103 bands covering the spectral range from 0.43 to 0.86 μm with an interval of 10 nm, and the image of each band measures 610 $\times$ 340 pixels with a spatial resolution of 1.3 m.

3) Pavia Center: The PC data set was obtained by the same Hyperion sensor of the PU data set with the same spatial resolution as the PU data set. However, it has one more band than the PU data set with the total number of bands as 103. The image of each band measures 1096×1096 pixels, which is much larger than the PU data set.

4) Botswana: The Botswana data set was obtained by the Hyperion sensor of the NASA EO-1 satellite over the Okavango Delta, Botswana, in 2001–2004. In Botswana, there are 242 bands in total that cover the spectral range from 0.4 to 2.5 μm with an interval of 10 nm. After removing the uncalibrated and noisy bands of water absorption features, the rest 145 bands are remained, and the image of each band measures 1476×256 pixels with a spatial resolution of 30 m.

5) Indian Pines: The IP data set was obtained by the AVIRIS sensor over the IP test site, Indiana. In this data set, there are 224 bands covering the spectral range from 0.4 to 2.5 μm. After removing the bands covering the region of water absorption, there are 200 bands remained, and the image of each band measures 145×145 pixels.

6) Washington DC Mall: The Washington DC Mall (WDCM) data set was obtained by the sensor of hyperspectral digital imagery collection experiment (HYDICE) over the National Mall in Washington, DC, in 1995. In WDCM, there are 210 bands in total that cover the wavelengths from 0.4 to 2.5 μm. After removing the bands of water vapor absorption, the remaining 191 bands are used in this article, and the image of each band measures 1280×307 pixels with a spatial resolution of 2.5 m.

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B. Evaluation Metrics

Four widely used metrics are employed to evaluate the performance of the proposed SSR-NET and comparison methods. Here, $\mathbf{R}_k(i, j)$ and $\mathbf{Z}_k(i, j)$ denote the element value at spatial location (i, j) in band k of the reference HR-HSI and the estimated HR-HSI.

1) Root-Mean-Squared Error: It can be used to measure the difference between $\mathbf{R}$ and $\mathbf{Z}$, which is defined as

$$\text{RMSE} = \sqrt{\frac{\sum_{k=1}^{L}\sum_{i=1}^{H}\sum_{j=1}^{W}(\mathbf{R}_k(i, j) - \mathbf{Z}_k(i, j))^2}{HWL}} \tag{21}$$

where the smaller the root-mean-squared error (RMSE), the better the performance.

2) Peak Signal-to-Noise Ratio: The peak SNR (PSNR) can evaluate the spatial quality of the reconstructed HR-HSI in unit of band. The PSNR of kth band is defined as

$$\text{PSNR} = 10\log_{10}\left(\frac{\max(\mathbf{R}_k)^2}{\frac{1}{HW}\|\mathbf{R}_k - \mathbf{Z}_k\|_2^2}\right) \tag{22}$$

where $\mathbf{R}_k$ and $\mathbf{Z}_k$, respectively, represent the reference image and the estimated image of the kth band. $\|\cdot\|_2$ refers to the two-norm. The final PSNR is the average of the PSNRs of all bands. The higher the PSNR, the better the performance.

3) Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS): The ERGAS [41] is specially designed for assessing the quality of high-resolution synthesized images, which measures the global statistical quality of the estimated HR-HSI. It is defined as

$$\text{ERGAS} = \frac{100}{r}\sqrt{\frac{1}{L}\sum_{k=1}^{L}\frac{\|\mathbf{R}_k - \mathbf{Z}_k\|_2^2}{\mu^2(\mathbf{R}_k)}} \tag{23}$$

where r refers to the ratio of the spatial downsampling ratio from HR-HSI to LR-HSI. $\mu(\mathbf{R}_k)$ denotes the mean value of the reference image of the kth band. The smaller the ERGAS, the better the performance.

4) Spectral Angle Mapper (SAM): The SAM [42] is generally utilized to evaluate the spectral information preservation degree at each pixel, which is defined as

$$\text{SAM} = \arccos\left(\frac{\langle\mathbf{R}(i, j), \mathbf{Z}(i, j)\rangle}{\|\mathbf{R}(i, j)\|_2\|\mathbf{Z}(i, j)\|_2}\right) \tag{24}$$

where $\mathbf{R}(i, j)$ and $\mathbf{Z}(i, j)$, respectively, denote the spectral vector of the reference and the estimated HR-HSI at the pixel position of (i, j). Besides, $\langle\mathbf{R}(i, j), \mathbf{Z}(i, j)\rangle$ refers to the inner product of $\mathbf{R}(i, j)$ and $\mathbf{Z}(i, j)$. The overall SAM is the average of the SAMs of all pixels. The lower the SAM, the better the performance.

C. Experimental Settings

For the IP data set that is limited by the spatial resolution, the center 64×64 subregion is cropped as the test image, and the rest region is used for training. More specifically, in each iteration, the training image with the same spatial resolution of 64×64 is randomly cropped from the training region. For all the other five data sets, the center 128×128 subregion is cropped as the test image, and the rest region is used for training. Similarly, in each iteration, the training image with the same spatial resolution of 128×128 is randomly cropped from the training region. It is worth mentioning that the training and test region are nonoverlapping, which is achieved by padding the test region with zeros in the data sets during the training stage. An example of the train and test regions of the Urban data set is illustrated in Fig. 4. The LR-HSI is downsampled with the ratio of r set to 4 from the blurred HR-HSI, which is in advance processed by 5×5 Gaussian filter with the standard deviation set to 2. The HR-MSI is composed of the five images that are located in HR-HSI at equal intervals.

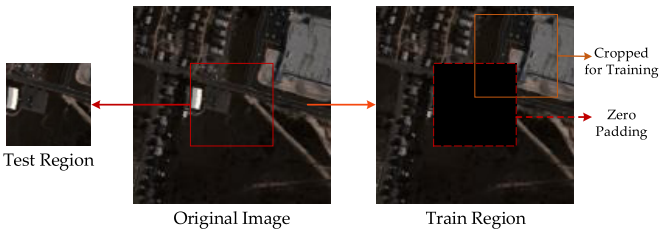


Fig. 4. Example of the train and test regions of the Urban data set. During the training stage, the test region in the data set is padded by zeros.

Two traditional methods of CNMF [12] and LTTR [25] and five deep learning methods of TFNet [37], ResTFNet [37], SSFCNN [38], ConSSFCNN [38], and MSDCNN [39] are selected as the comparison approaches to evaluate the performance of the proposed SSR-NET. For the traditional methods, except data processing, all the parameters are set as the same as the original literature. For all the deep learning models, the channel numbers of its input and output are adaptive to the used data set. During the training stage, Adam is selected as the optimizer. All the models are trained for 10 000 iterations with the learning rate of 1e-4. It is worth mentioning that since the training images are randomly cropped from the training area, in this article, iteration is used as the interval of metric measurement instead of the epoch.

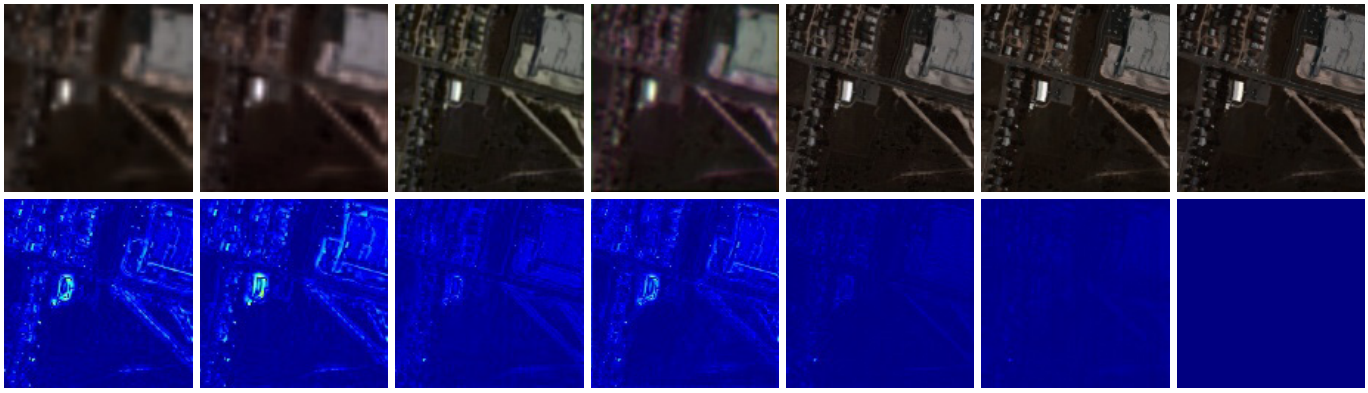
Besides, all the deep learning-based experiments are implemented by Pytorch 1.3.0 on Python 3.7. The computing equipment contains the 64-GB CPU memory and 1× GPU of GeForce GTX 1080Ti.

D. Ablation Study of SSR-NET

In order to explore the role of the components of SSR-NET in HSI and MSI fusion, some ablation experiments are conducted on the data set of Urban based on the five model variations of Spat-CNN, Spec-CNN, SpatR-NET, SpecR-NET, and SSR-NET.

The results of ablation experiments are shown in Table II and the fusion images of the model variations are illustrated in Fig. 5. Although SpatR-NET and SpecR-NET have a similar network architectures with Spat-CNN and Spec-CNN, the first two models perform quite better compared with the last two models. The reason is probably that the inputs of SpatR-NET and SpecR-NET are both HMSI that contains more information than a single LR-HSI or HR-MSI, and more valuable spatial/spectral feature can be learned under the guidance

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(a) (b) (c) (d) (e) (f) (g)

Fig. 5. Fusion results of Urban based on different model variations, where "GT" refers to the ground-truth image. The first row shows the R-G-B images (26-11-1 bands) of the estimated HR-HSIs, and the second row shows the difference images between the estimated R-G-B image and the reference R-G-B image (i.e., the ground-truth R-G-B image), which are processed by pseudocolor technique. (a) LR-HSI ↑. (b) Spat-CNN. (c) Spec-CNN. (d) patR-NET. (e) SpecR-NET. (f) SSR-NET. (g) GT.

TABLE II

ABLATION EXPERIMENTAL RESULTS OF THE PROPOSED SSR-NET ON THE URBAN DATA SET. THE ARROW ATTACHED TO THE METRICS POINTS TO THE BETTER PERFORMANCE. THE BEST SCORES ARE MARKED IN **BOLD**

Model	Urban			
	RMSE↓	PSNR↑	ERGAS↓	SAM↓
Spat-CNN	18.1819	20.6904	9.8345	11.5203
Spec-CNN	7.7957	28.0460	3.8693	7.8680
SpatR-NET	15.5343	22.0573	8.5497	9.9719
SpecR-NET	7.0083	28.9709	3.3848	5.1665
SSR-NET	**2.3693**	**38.3909**	**1.2173**	**2.3124**

of the spatial edge loss of $\mathcal{L}_{\text{spat}}$ and the spectral edge loss of $\mathcal{L}_{\text{spec}}$. When comparing Spec-CNN and SpecR-NET with Spat-CNN and SpatR-NET, it could be found that the first two models reconstructing the spectral information have a better performance than the last two models reconstructing the spatial information. Such results indicate that spectral information is easier to be reconstructed than spatial information. In other words, the spatial information is much more complicated than the spectral information. After combining the advantages of SpatR-NET and SpecR-NET, SSR-NET achieves the best image fusion performance, which is much better than each individual of them.

E. Comparison With State-of-the-Art Methods

To verify the effectiveness of the proposed SSR-NET, comparison experiments of the data sets of Urban, PU, PC, and Botswana are performed on seven state-of-the-art approaches, including CNMF [12], LTTR [25], TFNet [37], ResTFNet [37], SSFCNN [38], ConSSFCNN [38], and MSDCNN [39]. CNMF is a physical straightforward matrix factorization-based method, which aims to utilize the hyperspectral endmember matrix and the high-spatial-resolution abundance matrix to simulate the HR-HSI based on LR-HSI and HR-MSI. LTTR algorithm is a tensor-factorization-based method, which designs an LTTR prior to learning the relationships among spectral, spatial, and nonlocal modes, and correspondingly proposes a low tensor train (TT) rank (LTTR)-based HSI super-resolution approach. TFNet, ResTFNet, SSFCNN, ConSSFCNN, and MSDCNN belong to deep learning methods,

TABLE III

COMPARISON RESULTS OF THE PROPOSED SSR-NET AND SEVEN STATE-OF-THE-ART METHODS ON THE URBAN DATA SET. THE BEST SCORES ARE MARKED IN RED, AND THE SECOND SCORES ARE MARKED IN GREEN

Method	Urban			
	RMSE↓	PSNR↑	ERGAS↓	SAM↓
CNMF [12]	4.0198	36.0468	1.5612	2.4971
LTTR [25]	24.6837	20.2826	11.7672	9.1336
MSDCNN [39]	3.1317	35.9676	1.7652	3.1199
TFNet [37]	3.0405	36.2243	1.7146	2.8524
ResTFNet [37]	2.8916	36.6604	1.5980	2.7355
SSFCNN [38]	8.5801	27.2133	4.2196	8.7402
ConSSFCNN [38]	3.8841	34.0972	1.8671	3.1609
Our SSR-NET	2.3693	38.3909	1.2173	2.3124

TABLE IV

COMPARISON RESULTS OF THE PROPOSED SSR-NET AND SEVEN STATE-OF-THE-ART METHODS ON THE PU DATA SET. THE BEST SCORES ARE MARKED IN RED, AND THE SECOND SCORES ARE MARKED IN GREEN

Method	Pavia University			
	RMSE↓	PSNR↑	ERGAS↓	SAM↓
CNMF [12]	7.5816	30.5356	3.8939	2.2469
LTTR [25]	9.4202	28.6496	5.5504	5.1883
MSDCNN [39]	2.3207	40.5032	1.6358	2.6309
TFNet [37]	2.2427	40.8004	1.6053	2.4890
ResTFNet [37]	2.0578	41.5477	1.5068	2.3517
SSFCNN [38]	1.9410	42.0550	1.3957	2.2121
ConSSFCNN [38]	2.3916	40.2420	1.6340	2.4235
Our SSR-NET	1.6447	43.4938	1.2146	1.9329

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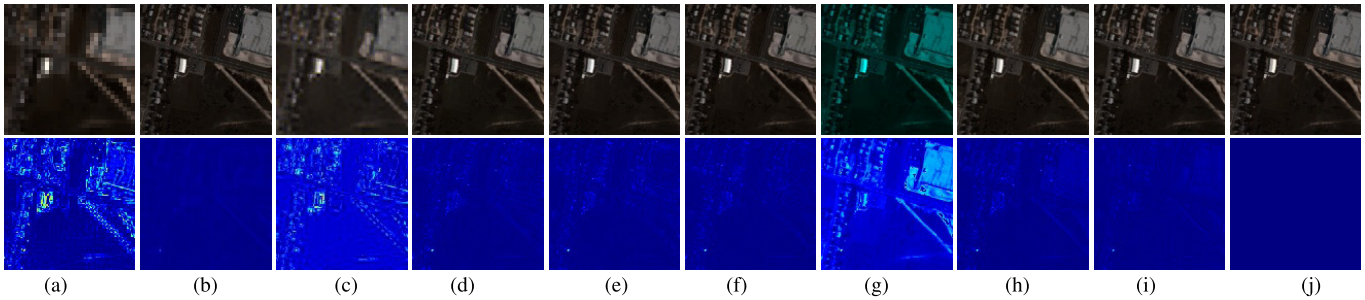


Fig. 6. Fusion results of different methods on Urban data set, where "ConSSF" and "GT' represent the ConSSFCNN and the ground-truth image. The first row shows the R-G-B images (26-11-1 bands) of the estimated HR-HSIs, and the second row shows the difference images between the estimated and the reference R-G-B images, which are processed by pseudocolor technique. (a) LR-HSI ↑. (b) CNMF. (c) LTTR. (d) MSDCNN. (e) TFNet. (f) ResTFNet. (g) SSFCNN. (h) ConSSF. (i) SSR-NET. (j) GT.

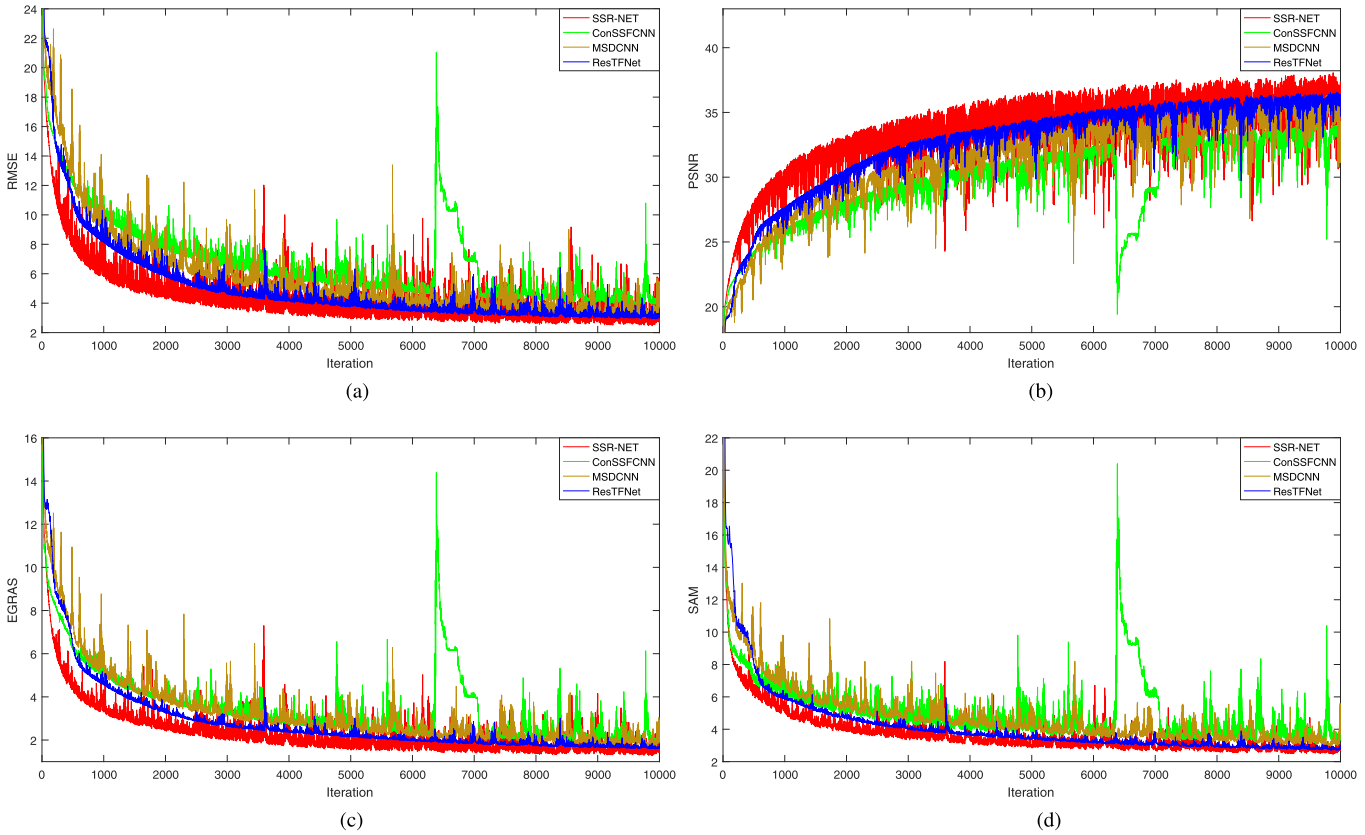


Fig. 7. Metrics between the proposed SSR-NET and three comparison methods of MSDCNN, ConSSFCNN, and ResTFNet on the Urban data set during the training stage. (a) RMSE. (b) PSNR. (c) ERGAS. (d) SAM.

but they have different emphases. TFNet and ResTFNet are two-stream networks that encode the spatial and spectral features independently and then decode the HR-HSI using the fusion of spatial and spectral features. Compared with TFNet, there are extra skip-connection operations in ResTFNet. SSFCNN and ConSSFCNN use the direct concatenated image of LR-HSI and HR-MSI to predict the HR-HSI, where the latter concatenates the HR-MSI in each convolutional layer. Based on residual learning and multiscale feature extraction, MSDCNN presents a multiscale and multidepth CNN for remote sensing image fusion.

1) Results on Urban: First, the comparison experiments based on the proposed SSR-NET and the aforementioned seven comparative methods are conducted on the Urban data set, and the experimental results are provided in Table III. In order to show the fusion performance intuitively, Fig. 6 illustrates the fusion performance of all the used methods with the fusion R-G-B images, which are selected from the estimated HR-HSI, and the difference images of the fusion R-G-B images and the corresponding reference R-G-B images. Among the deep learning methods, the performance of SSFCNN on Urban is not good enough, which falls far behind ConSSFCNN. It indicates that the direct concatenation between HR-MSI and LR-HSI↑ is not suitable for the Urban data set.

Fig. 7 shows all the four evaluation metrics of the proposed SSR-NET and comparative methods on the Urban data set during the training process. In the comparative

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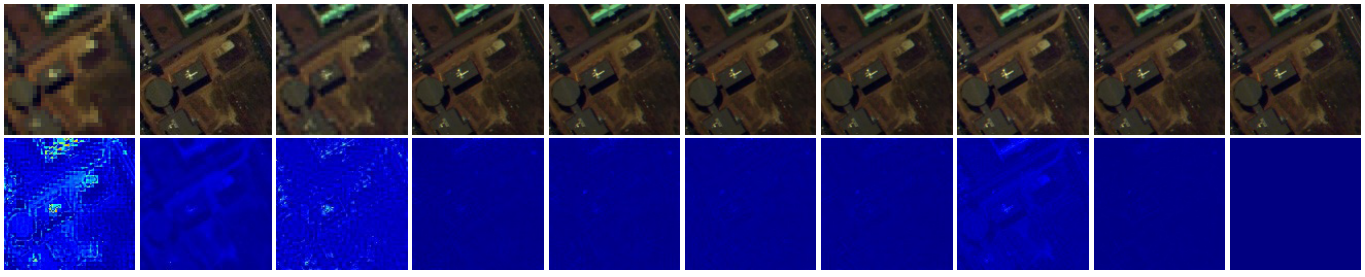


Fig. 8. Fusion results of different methods on PU data set, where "ConSSF" and "GT" respectively represent the ConSSFCNN and the ground-truth image. The first row shows the R-G-B images (67-29-1 bands) of the estimated HR-HSIs, and the second row shows the difference images between the estimated and the reference R-G-B images, which are processed by pseudocolor technique. (a) LR-HSI ↑. (b) CNMF. (c) LTTR. (d) MSDCNN. (e) TFNet. (f) ResTFNet. (g) SSFCNN. (h) ConSSF. (i) SSR-NET. (j) GT.

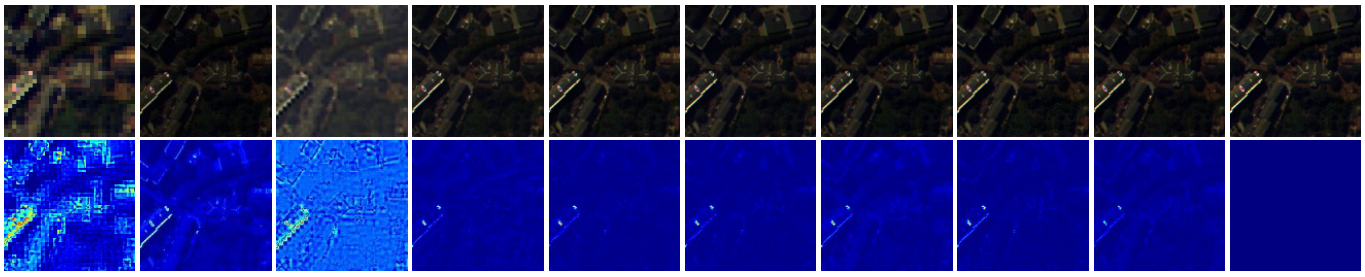


Fig. 9. Fusion results of different methods on PC data set, where "ConSSF" and "GT" represent the ConSSFCNN and the ground-truth image. The first row shows the R-G-B images (67-29-1 bands) of the estimated HR-HSIs, and the second row shows the difference images between the estimated and the reference R-G-B images, which are processed by pseudocolor technique. (a) LR-HSI ↑. (b) CNMF. (c) LTTR. (d) MSDCNN. (e) TFNet. (f) ResTFNet. (g) SSFCNN. (h) ConSSF. (i) SSR-NET. (j) GT.

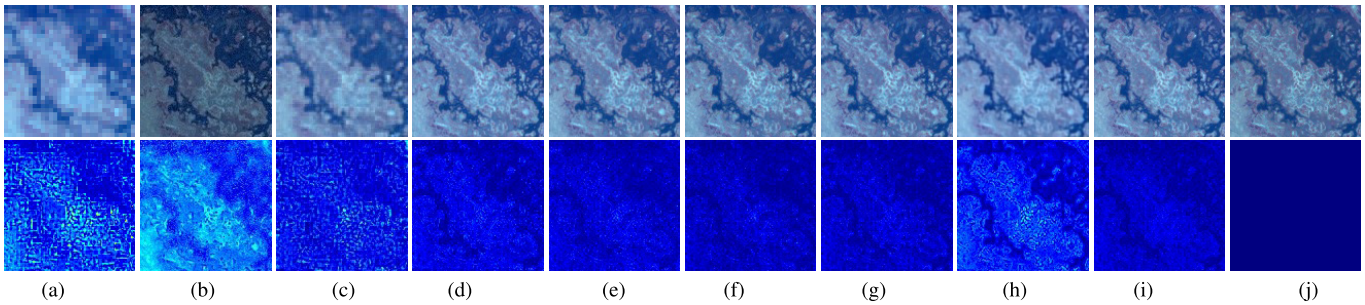


Fig. 10. Fusion results of different methods on Botswana data set, where "ConSSF" and "GT" represent the ConSSFCNN and the ground-truth image. The first row shows the R-G-B images (48-15-4 bands) of the estimated HR-HSIs, and the second row shows the difference images between the estimated and the reference R-G-B images, which are processed by pseudocolor technique. (a) LR-HSI ↑. (b) CNMF. (c) LTTR. (d) MSDCNN. (e) TFNet. (f) ResTFNet. (g) SSFCNN. (h) ConSSF. (i) SSR-NET. (j) GT.

methods, TFNet and SSFCNN are not shown in the figure due to the limit of space, and their better versions of ResTFNet and ConSSFCNN are enough for comparison. It can be seen that although the proposed SSR-NET has a larger fluctuation with noise, there is an obvious advantage of the convergence speed in SSR-NET when comparing with other methods, especially in the early training iterations from 0 to 5000. It mainly benefits from the proposed loss of $\mathcal{L}_{\text{spec}}$ and $\mathcal{L}_{\text{spat}}$. It can be also observed that our SSR-NET has the superior upper limits of performance on the used metrics, which can be found in the iterations from 5000 to 10 000.

2) Results on Pavia University: Second, Table IV shows the experimental results on the PU data set, and Fig. 8 shows the fusion R-G-B images and different images. In Table IV, it can be easily seen that for all four evaluation metrics, the proposed SSR-NET achieves the best performance among all the methods with the obvious advantage for PU data set. SSFCNN also obtains good performance on the PU data set. In contrast, ConSSFCNN performs worse than its simple version of SSFCNN, which may be caused by the unstable skip-concatenation that is harmful to training. ResTFNet performs better when comparing to TFNet, which is probably that the skip-connection in ResTFNet can extract more stable features based on stepwise residual strategy. According to Fig. 8, deep learning methods achieve superior fusion performance than traditional methods. By taking advantage of both spatial information in HR-MSI and spectral information in LR-HSI, the proposed SSR-NET can obtain the best fusion quality in a physical straightforward manner.

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Fig. 11. Fusion results of different methods on IP data set, where "ConSSF" and "GT" represent the ConSSFCNN and the ground-truth image. The first row shows the R-G-B images (29-15-4 bands) of the estimated HR-HSIs, and the second row shows the difference images between the estimated and the reference R-G-B images, which are processed by pseudocolor technique. (a) LR-HSI ↑. (b) CNMF. (c) LTTR. (d) MSDCNN. (e) TFNet. (f) ResTFNet. (g) SSFCNN. (h) ConSSF. (i) SSR-NET. (j) GT.

TABLE V

COMPARISON RESULTS OF THE PROPOSED SSR-NET AND SEVEN STATE-OF-THE-ART METHODS ON THE PC DATA SET. THE BEST SCORES ARE MARKED IN RED, AND THE SECOND SCORES ARE MARKED IN GREEN

Method	Pavia Center			
	RMSE↓	PSNR↑	ERGAS↓	SAM↓
CNMF [12]	13.9777	25.2221	11.4360	4.3635
LTTR [25]	28.3373	19.0836	31.1229	17.0800
MSDCNN [39]	4,1687	35.7307	4.6840	5.4004
TFNet [37]	4.2473	35.5686	4.8295	4.8437
ResTFNet [37]	3.9258	36.2522	4.4449	4.6073
SSFCNN [38]	4.7720	34.5569	5.7509	5.7962
ConSSFCNN [38]	5.4998	33.3239	6.1594	6.1196
Our SSR-NET	3.4102	37.4752	3.8964	3.8618

TABLE VI

COMPARISON RESULTS OF THE PROPOSED SSR-NET AND SEVEN STATE-OF-THE-ART METHODS ON THE BOTSWANA DATA SET. THE BEST SCORES ARE MARKED IN RED, AND THE SECOND SCORES ARE MARKED IN GREEN

Method	Botswana			
	RMSE↓	PSNR↑	ERGAS↓	SAM↓
CNMF [12]	26.3457	19.7166	9.4849	2.4866
LTTR [25]	9.1525	28.9000	14.2396	4.8682
MSDCNN [39]	0.5544	36.3497	3.1753	2.5996
TFNet [37]	0.4991	37.2622	2.7884	2.3161
ResTFNet [37]	0.4598	37.9749	2.6952	2.1537
SSFCNN [38]	1.4151	28.2108	11.4498	7.0424
ConSSFCNN [38]	2.5058	23.2475	15.2048	12.9631
Our SSR-NET	0.4106	38.9583	2.7994	2.0181

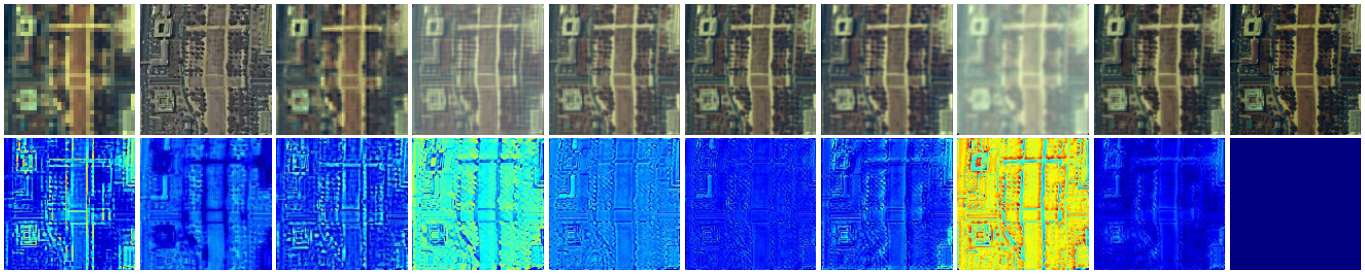
3) Results on Pavia Center: Table V lists the experimental results of our SSR-NET and the same comparative methods on the PC data set, and Fig. 9 illustrates the fusion performance with the fusion R-G-B images of our SSR-NET and the comparative methods. In Table V, it can be found that for all four evaluation metrics, the proposed SSR-NET obtains the superior performance than all the other comparative methods. Whether from the perspective of spatial reconstruction quality (ERGAS), spectral reconstruction quality (SAM), or the elementwise reconstruction quality (RMSE and PSNR), the proposed SSR-NET has an obvious advantage. Similar to the results of the PU data set, the deep learning methods are much better than traditional methods. SSFCCN also performs worse than ConSSFCNN, and ResTFNet still has better fusion quality than TFNet.

4) Results on Botswana: The experimental results of the Botswana data set on the proposed SSR-NET and the comparative methods are shown in Table VI, and the corresponding fusion R-G-B images are illustrated in Fig. 10. Among all the evaluation metrics, our SSR-NET obtains three best scores of RMSE, PSNR, and SAM and one competitive score of ERGAS. The first and second places of ERGAS are acquired by ResTFNet and TFNet. In the Botswana data set, the spatial resolution of a pixel is up to 30 m, and thus, its spatial information is much more complex than other data sets with a higher requirement for feature extraction. TFNet and ResTFNet, which are much deeper than our three-layer SSR-NET, have the advantage in extracting nonlinear deep feature that is beneficial for spatial reconstruction. It is probably that the proposed SSR-NET will perform better than TFNet and ResTFNet if SSR-NET has the same model size as TFNet and ResTFNet.

5) Results on Indian Pines: Table VII lists the comparison results of the IP data set between the proposed SSR-NET and the comparative methods, and Fig. 11 shows the corresponding fusion R-G-B images. It can be seen that our SSR-NET performs best among all the approaches in terms of RMSE, PSNR, and SAM. As for ERGAS, the proposed SSR-NET achieves comparative performance when comparing with other state-of-the-art methods. There are two potential factors leading to such a result. First, it is probably that the metric of ERGAS concentrates on global information. Instead, the proposed spatial edge loss and spectral edge loss stress more on the local features, ignoring the global features. Second, limited by the model size, SSR-NET has a disadvantage in a large receptive field.

6) Results on Washington DC Mall (WDCM): The results of the proposed SSR-NET and the seven comparative methods on the WDCM data set are shown in Table VIII, and the fusion results of the used approaches on this data set are illustrated in Fig. 12. It can be seen that similar to TFNet, the experimental results of our SSR-NET are in the second echelon, which just slightly falls behind ResTFNet. It is probably that compared with TFNet and ResTFNet, the model size of the proposed

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(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 12. Fusion results of different methods on Washington DC Mall data set, where "ConSSF" and "GT" represent the ConSSFCNN and the ground-truth image. The first row shows the R-G-B images (55-35-11 bands) of the estimated HR-HSIs, and the second row shows the difference images between the estimated and the reference R-G-B images, which are processed by pseudocolor technique. (a) LR-HSI ↑. (b) CNMF. (c) LTTR. (d) MSDCNN. (e) TFNet. (f) ResTFNet. (g) SSFCNN. (h) ConSSF. (i) SSR-NET. (j) GT.

TABLE VII

COMPARISON RESULTS OF THE PROPOSED SSR-NET AND SEVEN STATE-OF-THE-ART METHODS ON THE IP DATA SET. THE BEST SCORES ARE MARKED IN RED, AND THE SECOND SCORES ARE MARKED IN GREEN

Method	Indian Pines			
	RMSE↓	PSNR↑	ERGAS↓	SAM↓
CNMF [12]	10.2174	27.9440	3.2092	2.8280
LTTR [25]	44.1083	15.2404	226.8429	19.0087
MSDCNN [39]	4.2063	34.4827	2.5249	2.9217
TFNet [37]	4.2682	34.3559	2.1629	2.9586
ResTFNet [37]	3.9815	34.9598	2.0790	2.8160
SSFCNN [38]	12.0730	25.3245	12.2866	9.6967
ConSSFCNN [38]	7.4048	29.5705	15.5868	5.6114
Our SSR-NET	3.8475	35.2573	8.5339	2.7886

TABLE VIII

COMPARISON RESULTS OF THE PROPOSED SSR-NET AND SEVEN STATE-OF-THE-ART METHODS ON THE WASHINGTON DC MALL DATA SET. THE BEST SCORES ARE MARKED IN RED, AND THE SECOND SCORES ARE MARKED IN GREEN

Method	Washington DC Mall			
	RMSE↓	PSNR↑	ERGAS↓	SAM↓
CNMF [12]	57.7621	12.8979	1356.2909	32.6896
LTTR [25]	18.6020	22.7396	301.2300	13.3933
MSDCNN [39]	2.7328	36.7345	0.4778	0.8993
TFNet [37]	1.9704	39.5756	0.3460	0.6756
ResTFNet [37]	1.7348	40.6816	0.3035	0.5836
SSFCNN [38]	16.4613	21.1373	3.1689	6.8978
ConSSFCNN [38]	14.0913	22.4876	2.6805	5.8734
Our SSR-NET	2.0591	39.1928	0.3603	0.7201

SSR-NET is rather smaller, which limits the performance of our model on spatial feature extraction of a large receptive field. It is also our future work in HSI and MSI fusion.

Overall, the proposed SSR-NET has the best fusion performance on the five data sets, including Urban, PU, PC, Botswana, and IP, and achieves the comparable results in the WDCM data set. Traditional methods of CNMF and LTTR perform worse than deep learning methods. Among deep learning methods, compared with SSFCNN and ConSSFCNN, our SSR-NET has a more stable performance. Compared with TFNet, ResTFNet, and MSDCNN, SSR-NET has not only better performance but also a much smaller model size.

TABLE IX

METRIC SCORES MEASURED BETWEEN THE REFERENCE HR-HSI AND THE OUTPUTS OF DIFFERENT STAGES IN SSR-NET. THE BEST SCORES ARE MARKED IN **BOLD**

Output	Urban			
	RMSE↓	PSNR↑	ERGAS↓	SAM↓
HMSI	14.5836	22.6058	7.9786	7.0940
Spectral HR-HSI	2.5869	37.6276	1.3295	2.6184
Spatial HR-HSI	**2.3693**	**38.3909**	**1.2173**	**2.3124**

F. Analysis of the Interpretability of SSR-NET

Although deep learning methods obtain excellent performance, as black-box models, they are usually lack of interpretability. Different from the existing deep learning methods used in LR-HSI and HR-MSI fusion, the proposed SSR-NET is interpretable to some extent. Due to the CMMI, all the outputs of different stages in SSR-NET, which have the same size as the reference HR-HSI, can serve as the estimated HR-HSI with different qualities. To explore the process of the fusion HR-HSI in the proposed SSR-NET, the outputs of different stages of the Urban data set are quantitatively measured by the above four evaluation metrics. The experimental results are shown in Table IX.

In the table, HMSI denotes the concatenation of LR-HSI and HR-MSI according to (4). Spectral HR-HSI and spatial HR-HSI represent the output of SpecRN and SpatRN, respectively. It could be found that the fusion qualities of HMSI, spectral HR-HSI, and spatial HR-HSI are increasing, which demonstrates that the proposed SSR-NET is a physical straightforward model.

G. Analysis of Model Complexity

For deep learning methods, model parameters and test time are also important indicators for performance evaluation. In this subsection, the number of model parameters, its floating-point operations (FLOPs), and its test time of the proposed SSR-NET and the other five deep learning methods is compared quantitatively, and the detailed data are provided

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TABLE X

MODEL SIZE, FLOPS, AND TEST TIME OF THE PROPOSED SSR-NET AND THE OTHER DEEP LEARNING MODELS. THE BEST DATA ARE MARKED IN **BOLD**

Model	Urban		
	Params (M)	FLOPs (G)	Test Time (Ms)
MSDCNN [39]	1.82	59.76	438
TFNet [37]	2.50	18.3	215
ResTFNet [37]	2.38	**17.04**	206
SSFCNN [38]	1.14	37.24	232
ConSSFCNN [38]	1.16	38.16	268
Our SSR-NET	**0.69**	23.26	**151**

in Table X. The experiments are still performed on the test area of the Urban data set using only CPU.

According to the results, it is easily seen that the proposed SSR-NET has an obvious advantage in both model parameters and test time in comparison with the other deep learning models. Compared with the TFNet, the parameters of our SSR-NET are only 28% of TFNet, but the test speed is about 50% faster than it. As for FLOPs, the performance of our SSR-NET just slightly falls behind TFNet and ResTFNet. It is probably that all layers of our SSR-NET are operated at the pixel level. Taking the fusion quality into consideration again, the proposed SSR-NET has the overall advantage in the field of LR-HSI and HR-MSI fusion.

V. CONCLUSION

In this article, an interpretable CNN-based framework of SSR-NET is proposed for HSI and MSI fusion. The proposed SSR-NET contains three components of CMMI, SpatRN, and SpecRN. SpatRN and SpecRN are, respectively, optimized by two kinds of losses of spatial edge loss ($\mathcal{L}_{\text{spat}}$) and spectral edge loss ($\mathcal{L}_{\text{spec}}$), which are specially designed for the spatial and spectral restorations. Comparative experiments of the proposed SSR-NET and some state-of-the-art methods are conducted on six widely used hyperspectral data sets, including Urban, PU, PC, and Botswana. The superior experimental results of SSR-NET demonstrate the effectiveness of the proposed method. Besides the scores of evaluation metrics, our SSR-NET also has an obvious advantage in model size and test time in the type of deep learning methods.

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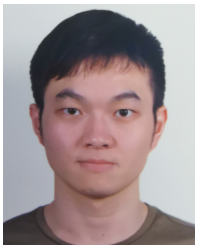
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